R Code

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2026-03-29

My Basic R Code

```r
#Total paper
#Preliminaries
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v readr     2.1.6
## v ggplot2   4.0.2     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.3.1
## v purrr     1.2.1     v tidyr     1.3.2

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Preliminary

```r
setwd("C:/Users/Owner/Documents/BallState/DSCI-609 SocSci/Paper")
gss_data <- read.csv("gss_data.csv")

# Keep columns where the count of non-NA values is greater than 0
```

```r
gss_data1 <- gss_data[, colSums(is.na(gss_data)) < nrow(gss_data)]

gss_data1$sex <- as_factor(gss_data1$sex)

# Cleaning missing values before analysis [cite: 69]
gss_data1$happy[gss_data1$happy > 3] <- NA
gss_data1$goodlife[gss_data1$goodlife > 5] <- NA
gss_data1$satfin[gss_data1$satfin > 6000] <- NA
gss_data1$sex[gss_data1$sex > 2] <- NA
```

```
## Warning in Ops.factor(gss_data1$sex, 2): '>' not meaningful for factors
```

```r
gss_data1$degree[gss_data1$degree > 4] <- NA
gss_data1$realinc[gss_data1$realinc > 100000] <- NA
gss_data1$age[gss_data1$age > 90] <- NA
gss_data1$polviews[gss_data1$polviews > 7] <- NA

# Narrow the scope to the 3-5 key variables suggested in the approach [cite: 36, 37]
gss_data2 <- gss_data1[, c("happy", "goodlife", "satfin", "degree", "realinc", "age", "sex", "polviews")
```

Batch 1

```r
#Batch1
library(tidyverse)
library(car) # For Levene's Test
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:purrr':
##
##     some

## The following object is masked from 'package:dplyr':
##
##     recode
```

```r
library(ggplot2)

# 1. Precise Cleaning for GSS codes
# We must ensure 'sex' is a factor and 'goodlife' is numeric and filtered
gss_clean <- gss_data2 %>%
  mutate(
    # Convert goodlife to numeric and handle GSS missing codes (0, 8, 9) [cite: 68-72]
    goodlife_num = as.numeric(as.character(goodlife)),
    goodlife_num = ifelse(goodlife_num > 5, NA, goodlife_num),
```

```r
    # Force sex to be a Factor (this fixes the Levene's Test error)
    sex_factor = as.factor(sex)
  ) %>%
  # Filter to keep ONLY Male and Female and remove any remaining NAs
  filter(sex_factor %in% c("male", "female"), !is.na(goodlife_num)) %>%
  # Drop unused levels like 'skipped on web' or 'iap'
  mutate(sex_factor = fct_drop(sex_factor))

# 2. Verify the Descriptive Statistics
# You should see actual numbers for Mean and SD now
gss_clean %>%
  group_by(sex_factor) %>%
  summarise(
    Mean = mean(goodlife_num),
    SD = sd(goodlife_num),
    N = n()
  )
```

```
## # A tibble: 2 x 4
##   sex_factor  Mean    SD     N
##   <fct>      <dbl> <dbl> <int>
## 1 female      2.58  1.10 15934
## 2 male        2.43  1.07 12796
```

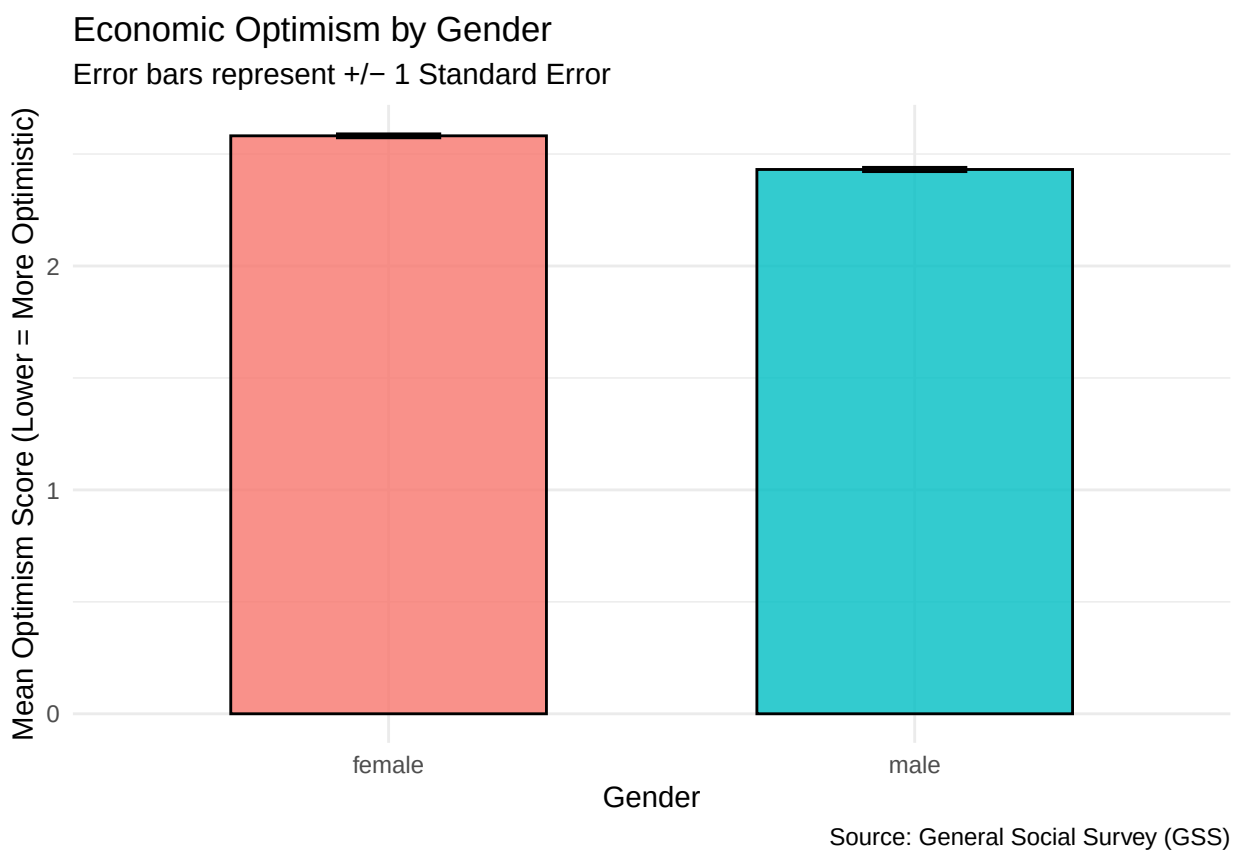
```r
# 3. Homogeneity of Variance (Levene's Test)
# This will now run because sex_factor is an explicit factor
lt <- leveneTest(goodlife_num ~ sex_factor, data = gss_clean)
print(lt)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##          Df F value    Pr(>F)
## group     1  71.722 < 2.2e-16 ***
##       28728
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 4. Independent Samples t-test
# We compare the means of these two groups [cite: 91, 99]
t_test_results <- t.test(goodlife_num ~ sex_factor, data = gss_clean, var.equal = TRUE)

print(t_test_results)
```

```
##
##  Two Sample t-test
##
## data:  goodlife_num by sex_factor
## t = 11.645, df = 28728, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group female and group male is not equal to
## 95 percent confidence interval:
##  0.1249716 0.1755550
## sample estimates:
## mean in group female   mean in group male
##             2.581336             2.431072
```

```r
# 5. Prepare the plotting data (Calculating Standard Error)
batch1_plot_data <- gss_clean %>%
  group_by(sex_factor) %>%
  summarise(
    Mean = mean(goodlife_num, na.rm = TRUE),
    SD = sd(goodlife_num, na.rm = TRUE),
    N = n(),
    SE = SD / sqrt(N)  # Formula for Standard Error
  )

# 6. Create the Bar Chart
ggplot(batch1_plot_data, aes(x = sex_factor, y = Mean, fill = sex_factor)) +
  geom_bar(stat = "identity", width = 0.6, color = "black", alpha = 0.8) +
  geom_errorbar(aes(ymin = Mean - SE, ymax = Mean + SE), width = 0.2) +
  labs(
    title = "Economic Optimism by Gender",
    subtitle = "Error bars represent +/- 1 Standard Error",
    x = "Gender",
    y = "Mean Optimism Score (Lower = More Optimistic)",
    caption = "Source: General Social Survey (GSS)"
  ) +
  theme_minimal() +
  scale_fill_manual(values = c("female" = "#f8766d", "male" = "#00bfc4")) +
  theme(legend.position = "none") # Hide legend as X-axis labels are clear
```



Batch 2

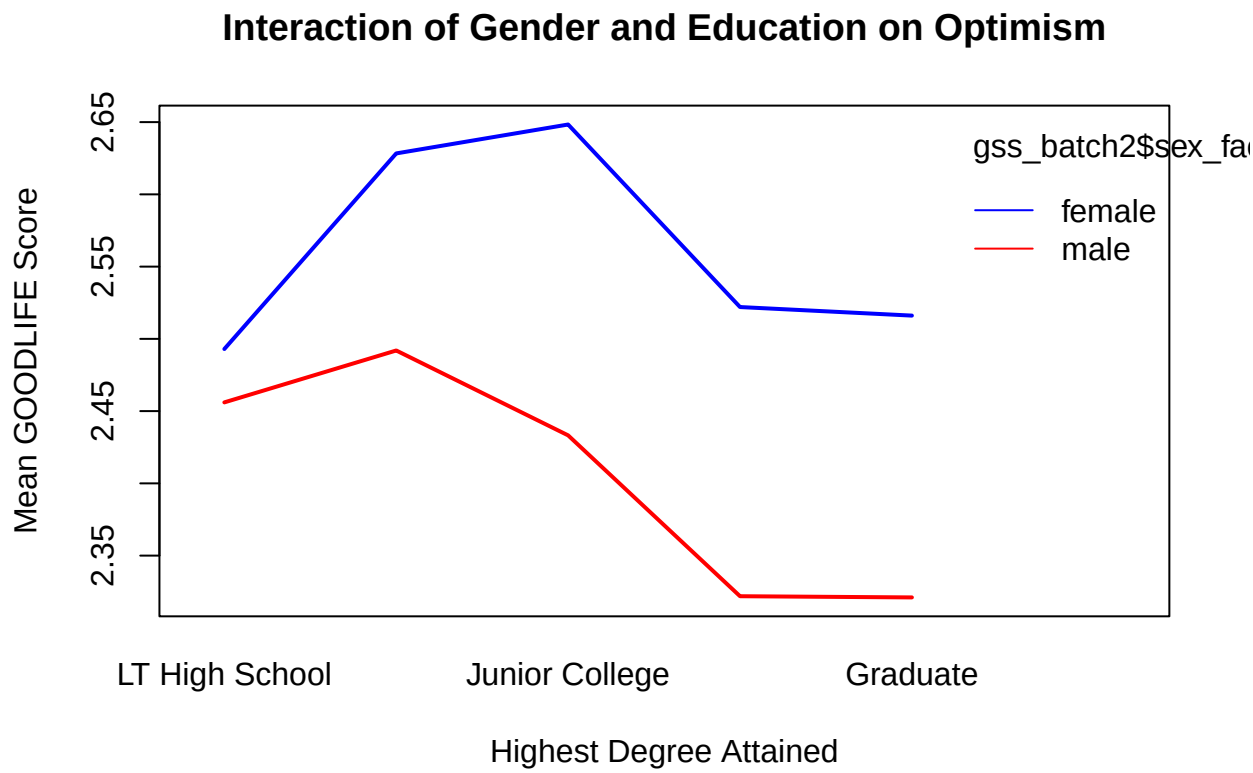
```r
#Batch2
# --- Batch 2: Two-Way ANOVA ---
# IV1: sex_factor, IV2: degree, DV: goodlife_num

# 1. Prepare the Data
gss_batch2 <- gss_clean %>%
  mutate(
    # Ensure 'degree' is a factor with correct GSS labels [cite: 57]
    degree_factor = as.factor(degree),
    degree_factor = fct_recode(degree_factor,
                               "LT High School" = "0",
                               "High School" = "1",
                               "Junior College" = "2",
                               "Bachelor" = "3",
                               "Graduate" = "4"
    )
  ) %>%
  filter(!is.na(degree_factor))

# Generate the Table Values for Batch 2
ipv <-gss_batch2 %>%
  group_by(degree_factor) %>%
  summarise(
    Mean_Optimism = mean(goodlife_num, na.rm = TRUE),
    Std_Dev = sd(goodlife_num, na.rm = TRUE),
    N = n()
  )
print(ipv)
```

```
## # A tibble: 5 x 4
##   degree_factor  Mean_Optimism Std_Dev     N
##   <fct>                  <dbl>   <dbl> <int>
## 1 LT High School          2.48    1.11  3783
## 2 High School             2.57    1.10 14243
## 3 Junior College          2.56    1.11  2236
## 4 Bachelor                2.43    1.06  5385
## 5 Graduate                2.43    1.03  3030
```

```r
# 2. Run the Two-Way ANOVA (Interaction Model) [cite: 122-125]
anova_results <- aov(goodlife_num ~ sex_factor * degree_factor, data = gss_batch2)

# 3. View the ANOVA Table
summary(anova_results)
```

```
##                              Df Sum Sq Mean Sq F value  Pr(>F)
## sex_factor                    1    159  158.85 134.907 < 2e-16 ***
## degree_factor                 4    109   27.19  23.096 < 2e-16 ***
## sex_factor:degree_factor      4     20    4.92   4.175 0.00222 **
## Residuals                 28667  33754    1.18
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 4. Interaction Plot (Visualizing the "Glass Ceiling") [cite: 126-129]
interaction.plot(gss_batch2$degree_factor, gss_batch2$sex_factor, gss_batch2$goodlife_num,
                 xlab = "Highest Degree Attained",
                 ylab = "Mean GOODLIFE Score",
                 main = "Interaction of Gender and Education on Optimism",
                 col = c("blue", "red"), lty = 1, lwd = 2)
```



Preliminary Part 2

```r
library(WDI)
# Pull with a more stable GDP code:
# NY.GDP.PCAP.PP.CD is GDP per capita, PPP (current international $)
world_data <- WDI(
  country = "all",
  indicator = c(
    Life_Expectancy = "SP.DYN.LE00.IN",
    GDP_per_capita  = "NY.GDP.PCAP.PP.CD",
    Health_Spend    = "SH.XPD.CHEX.GD.ZS",
    Education_Rate  = "SE.SEC.ENRR"
  ),
  start = 2005,
  end = 2022,
  extra = FALSE
```

```r
)

#Added to get regions
#Access the built-in country metadata
country_info <- as.data.frame(WDI_data$country)

# Select only the columns we need: ISO code and Region
region_map <- country_info[, c("iso2c", "region")]

# Merge it into your world_data
world_data_with_regions <- merge(world_data, region_map, by = "iso2c", all.x = TRUE)

# Filter for rows that are actual countries (removes aggregate regions like 'Arab World')
world_data1 <- world_data_with_regions[world_data_with_regions$region != "Aggregates", ]

# Filter for rows that are actual countries (removes aggregate regions like 'Arab World')
world_data2 <- world_data1[world_data1$region != "Aggregates", ]

# Define the columns you want to get rid of
unwanted_cols <- c("iso2c")

world_data3 <- world_data2 %>%
  # Drop only the columns that actually exist in the data
  dplyr::select(-one_of(intersect(names(.), unwanted_cols))) %>%

  # Rename country to country_name
  rename(country_name = country)
```

Batch 3

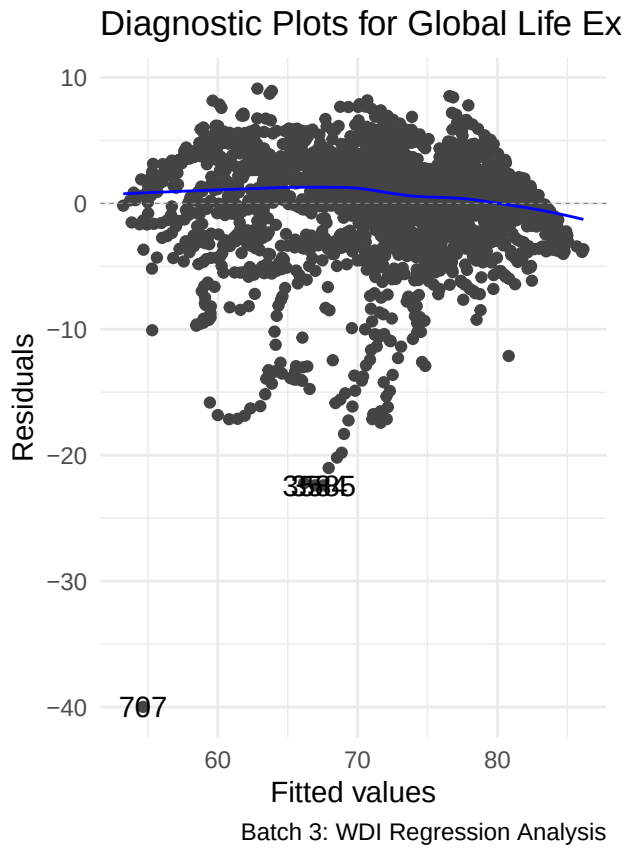
```r
#Batch3
# --- Batch 3: Multiple Regression ---
# Dataset: WDI
# DV: Life_Expectancy
# IVs: GDP_per_capita, Health_Spend, Education_Rate
library(ggfortify)

# 1. Run the Regression Model
# Using log() for GDP as recommended in the Paper Approach [cite: 170]
regression_model <- lm(Life_Expectancy ~ log(GDP_per_capita) + Health_Spend + Education_Rate,
                       data = world_data3)

# 2. View the Results
summary(regression_model)
```

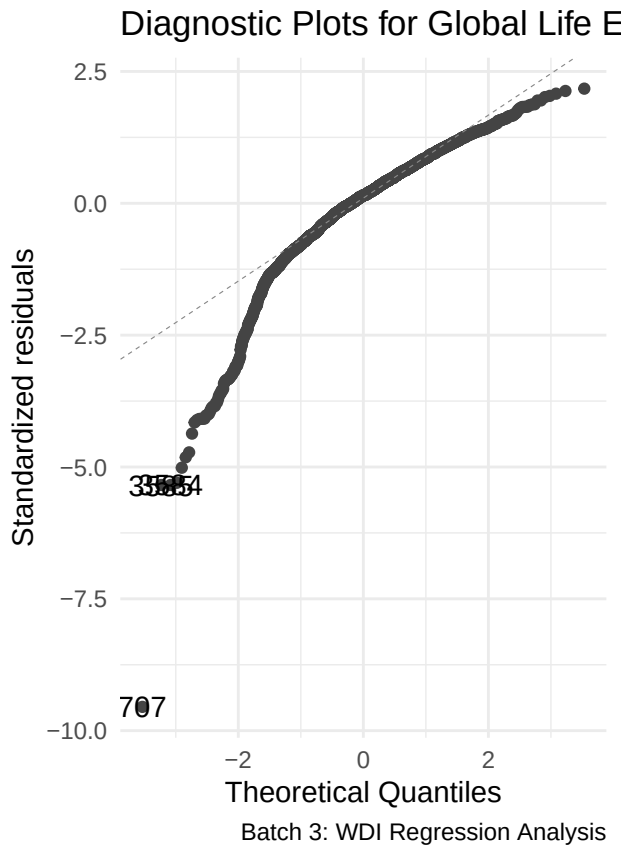
```
##
## Call:
## lm(formula = Life_Expectancy ~ log(GDP_per_capita) + Health_Spend +
##     Education_Rate, data = world_data3)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -39.989  -1.811   0.623   2.637   9.113
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         24.401302   0.879255  27.752  < 2e-16 ***
## log(GDP_per_capita)  4.212076   0.123502  34.105  < 2e-16 ***
## Health_Spend         0.154025   0.033048   4.661 3.32e-06 ***
## Education_Rate       0.082287   0.005361  15.348  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.195 on 2436 degrees of freedom
##   (1466 observations deleted due to missingness)
## Multiple R-squared:  0.7319, Adjusted R-squared:  0.7316
## F-statistic:  2217 on 3 and 2436 DF,  p-value: < 2.2e-16
```

```r
# Create both plots side-by-side
autoplot(regression_model, which = 1:2, ncol = 2) +
  theme_minimal() +
  labs(title = "Diagnostic Plots for Global Life Expectancy Model",
       caption = "Batch 3: WDI Regression Analysis")
```

```
## Warning: `fortify(<lm>)` was deprecated in ggplot2 4.0.0.
## i Please use `broom::augment(<lm>)` instead.
## i The deprecated feature was likely used in the ggfortify package.
##   Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the ggfortify package.
##   Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## i The deprecated feature was likely used in the ggfortify package.
##   Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```





Batch 4

```
#Batch4
library(mediation)
```

```
## Loading required package: MASS

##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select

## Loading required package: Matrix

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
## Loading required package: mvtnorm

## Loading required package: sandwich

## Registered S3 method overwritten by 'lme4':
##   method           from
##   na.action.merMod car

## mediation: Causal Mediation Analysis
## Version: 4.5.1
```

```r
library(DiagrammeR)

# 1. Clean and Prepare GSS Data
gss_med <- gss_data2 %>%
  mutate(
    # Reverse code happy: Higher = Happier (3 = Very Happy, 1 = Not Too Happy)
    happy_rev = case_when(
      happy == 1 ~ 3,
      happy == 2 ~ 2,
      happy == 3 ~ 1,
      TRUE ~ NA_real_
    ),
    # Ensure income and degree are numeric
    income_num = as.numeric(as.character(realinc)),
    edu_num = as.numeric(as.character(degree))
  ) %>%
  drop_na(happy_rev, income_num, edu_num)

# 2. Path A: Education -> Income
fit.mediator <- lm(income_num ~ edu_num, data = gss_med)

# 3. Path B & C': Education + Income -> Happiness
fit.dv <- lm(happy_rev ~ edu_num + income_num, data = gss_med)

# 4. Run the Mediation Analysis
med_results <- mediate(fit.mediator, fit.dv, treat = "edu_num", mediator = "income_num", boot = TRUE, s
```

```
## Running nonparametric bootstrap
```

```r
# 5. Summary
summary(med_results)
```

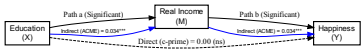
```
##
## Causal Mediation Analysis
##
## Nonparametric Bootstrap Confidence Intervals with the Percentile Method
##
##                    Estimate 95% CI Lower 95% CI Upper p-value
## ACME            0.03366821   0.03184882   0.03529992  <2e-16 ***
## ADE             0.00018208  -0.00398599   0.00434914    0.92
## Total Effect    0.03385029   0.02928831   0.03763689  <2e-16 ***
```

```
## Prop. Mediated  0.99462096   0.88455967   1.13593432  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Sample Size Used: 59727
##
##
## Simulations: 100
```

```r
# Path a (Education -> Income)
model_a <- lm(scale(income_num) ~ scale(edu_num), data = gss_med)
beta_a <- round(coef(model_a)[2], 3)

# Path b & c' (Income and Education -> Happiness)
model_bc <- lm(scale(happy_rev) ~ scale(edu_num) + scale(income_num), data = gss_med)
beta_c_prime <- round(coef(model_bc)[2], 3) # Direct effect
beta_b <- round(coef(model_bc)[3], 3)       # Mediator effect

# Print them to use in the diagram
cat("Path a:", beta_a, "\nPath b:", beta_b, "\nPath c':", beta_c_prime)
```

```
## Path a: 0.336
## Path b: 0.182
## Path c': 0
```

```r
library(DiagrammeR)

grViz("
  digraph mediation {
    # General Graph Settings
    graph [layout = dot, rankdir = LR]
    node [shape = box, fontname = Helvetica, penwidth = 2]

    # Define the Nodes (X, M, Y)
    X [label = 'Education\n(X)']
    M [label = 'Real Income\n(M)']
    Y [label = 'Happiness\n(Y)']

    # Define the Paths (Arrows)
    # Using your actual Batch 4 results:
    X -> M [label = 'Path a (Significant)']
    M -> Y [label = 'Path b (Significant)']

    # Path c-prime is the 'ADE' from your PDF (0.00018, p = 0.94)
    # We use a dashed line because it is NOT significant
    X -> Y [label = 'Direct (c-prime) = 0.00 (ns)', style = dashed]

    # Add the Total Effect and Indirect Effect as a note
    edge [fontname = Helvetica, fontsize = 10]

    # Optional: Label the indirect effect (ACME)
    X -> M -> Y [color = blue, label = 'Indirect (ACME) = 0.034***']
  }
")
```



Batch 5

```
#Batch5
library(tidyverse)
library(lme4)
library(performance)
library(sjPlot)
```

```
##
## Attaching package: 'sjPlot'

## The following object is masked from 'package:ggplot2':
##
##     set_theme
```

```
library(ggplot2)

setwd("C:/Users/Owner/Documents/BallState/DSCI-609 SocSci/Paper")

#Load the raw data
oecd_raw <- read.csv("oced_bli.csv")

#Create the Regional Mapping
# We must ensure North America and Sub-Saharan Africa are distinct groups
region_map <- data.frame(
  country_name = c(
    "Australia", "Austria", "Belgium", "Brazil", "Canada", "Chile", "Colombia",
```

```r
    "Costa Rica", "Czechia", "Denmark", "Estonia", "Finland", "France", "Germany",
    "Greece", "Hungary", "Iceland", "Ireland", "Israel", "Italy", "Japan", "Korea",
    "Latvia", "Lithuania", "Luxembourg", "Mexico", "Netherlands", "New Zealand",
    "Norway", "Poland", "Portugal", "Russia", "Slovak Republic", "Slovenia",
    "South Africa", "Spain", "Sweden", "Switzerland", "Türkiye", "United Kingdom",
    "United States"
  ),
  region = c(
    "East Asia & Pacific", "Europe & Central Asia", "Europe & Central Asia",
    "Latin America & Caribbean", "North America", "Latin America & Caribbean",
    "Latin America & Caribbean", "Latin America & Caribbean", "Europe & Central Asia",
    "Europe & Central Asia", "Europe & Central Asia", "Europe & Central Asia",
    "Europe & Central Asia", "Europe & Central Asia", "Europe & Central Asia",
    "Europe & Central Asia", "Europe & Central Asia", "Europe & Central Asia",
    "Middle East & North Africa", "Europe & Central Asia", "East Asia & Pacific",
    "East Asia & Pacific", "Europe & Central Asia", "Europe & Central Asia",
    "Europe & Central Asia", "Latin America & Caribbean", "Europe & Central Asia",
    "East Asia & Pacific", "Europe & Central Asia", "Europe & Central Asia",
    "Europe & Central Asia", "Europe & Central Asia", "Europe & Central Asia",
    "Europe & Central Asia", "Sub-Saharan Africa", "Europe & Central Asia",
    "Europe & Central Asia", "Europe & Central Asia", "Europe & Central Asia",
    "Europe & Central Asia", "North America"
  )
)

#Clean and Pivot the Data
batch_5_data <- oecd_raw %>%
  dplyr::filter(
    Indicator %in% c(
      "Life satisfaction",
      "Quality of support network",
      "Employment rate",
      "Household net adjusted disposable income",
      "Housing expenditure"
    ),
    Inequality == "Total"
  ) %>%
  dplyr::select(Country, Indicator, OBS_VALUE) %>%
  tidyr::pivot_wider(names_from = Indicator, values_from = OBS_VALUE) %>%
  dplyr::rename(
    country_name = Country,
    life_satisfaction = `Life satisfaction`,
    social_support = `Quality of support network`,
    employment_rate = `Employment rate`,
    GDP_per_capita = `Household net adjusted disposable income`,
    housing_affordability = `Housing expenditure`
  ) %>%
  dplyr::left_join(region_map, by = "country_name") %>%
  # Crucial: This is where your previous code stopped/broke.
  # We drop rows with missing values to ensure the Factor Analysis has data to work with.
  drop_na(life_satisfaction, social_support, employment_rate, GDP_per_capita, housing_affordability, re

#Fit a Null Multilevel Model
```

```r
mlm_null <- lmer(life_satisfaction ~ 1 + (1 | region), data = batch_5_data)
cat("Null Model ICC (Regional variance focus):\n")
```

```
## Null Model ICC (Regional variance focus):
```

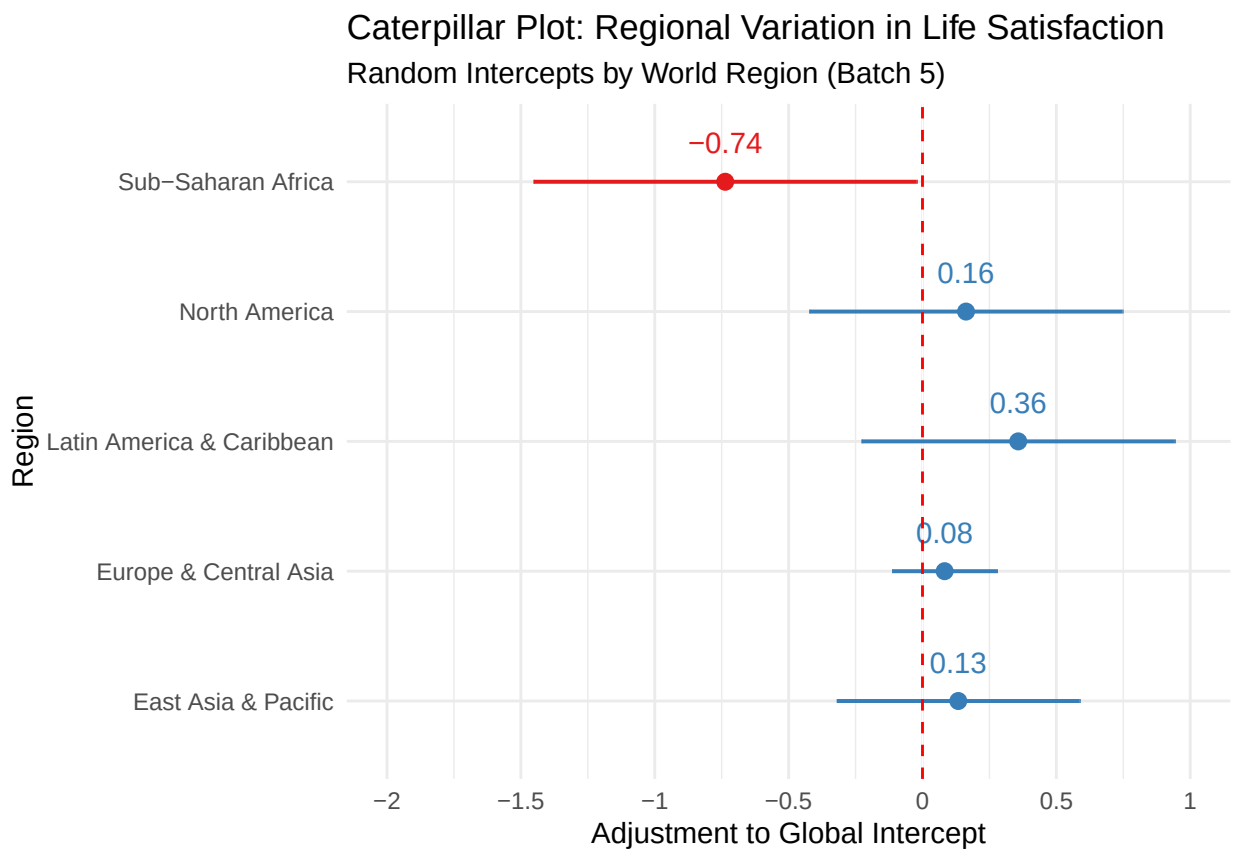
```r
print(performance::icc(mlm_null))
```

```
## # Intraclass Correlation Coefficient
##
##     Adjusted ICC: 0.330
##   Unadjusted ICC: 0.330
```

```r
#Fit the final requested model
mlm_final <- lmer(life_satisfaction ~ social_support + (1 | region), data = batch_5_data)
summary(mlm_final)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: life_satisfaction ~ social_support + (1 | region)
##    Data: batch_5_data
##
## REML criterion at convergence: 65.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.86922 -0.84873  0.05895  0.52786  1.65523
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  region   (Intercept) 0.2642   0.5140
##  Residual             0.2690   0.5186
## Number of obs: 35, groups:  region, 5
##
## Fixed effects:
##                Estimate Std. Error t value
## (Intercept)    -0.88743    1.88731  -0.470
## social_support  0.08173    0.02098   3.896
##
## Correlation of Fixed Effects:
##             (Intr)
## socil_spprt -0.990
```

```r
# Create the Caterpillar Plot for the final model
plot_model(mlm_final, type = "re", show.values = TRUE, value.offset = .3) +
  theme_minimal() +
  labs(
    title = "Caterpillar Plot: Regional Variation in Life Satisfaction",
    subtitle = "Random Intercepts by World Region (Batch 5)",
    y = "Adjustment to Global Intercept",
    x = "Region"
  ) +
  # Adds a horizontal line at 0 (the global average)
  geom_hline(yintercept = 0, linetype = "dashed", color = "red")
```



Batch 6

```
#Batch6
library(tidyverse)
library(psych)
```

```
##
## Attaching package: 'psych'

## The following object is masked from 'package:mediation':
##
##     mediate

## The following object is masked from 'package:car':
##
##     logit

## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```r
#Select the indicators for the Quality of Life Factor
# Now we select the numeric columns from our clean batch_5_data
efa_data <- batch_5_data %>%
  dplyr::select(
    GDP_per_capita,
    housing_affordability,
    employment_rate,
    social_support,
    life_satisfaction
  )

#Suitability tests
print(KMO(cor(efa_data)))
```

```
## Kaiser-Meyer-Olkin factor adequacy
## Call: KMO(r = cor(efa_data))
## Overall MSA =  0.74
## MSA for each item =
##        GDP_per_capita housing_affordability       employment_rate
##                  0.69                  0.68                  0.86
##        social_support     life_satisfaction
##                  0.84                  0.69
```

```r
print(bartlett.test(efa_data))
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  efa_data
## Bartlett's K-squared = 1888.1, df = 4, p-value < 2.2e-16
```

```r
# 1. Run the Parallel Analysis and Scree Plot
# n.iter = 100 runs a simulation to show where 'random' factors would fall
parallel_results <- fa.parallel(
  efa_data,
  fm = "minres", # Minimum Residual factor method
  fa = "fa",     # Looking specifically for Factors
  n.iter = 100,
  main = "Scree Plot: Identifying the 'Quality of Life' Factor"
)
```

```
## Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
## ultra-Heywood case was detected.  Examine the results carefully
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## Parallel analysis suggests that the number of factors =  1  and the number of components =  NA
```
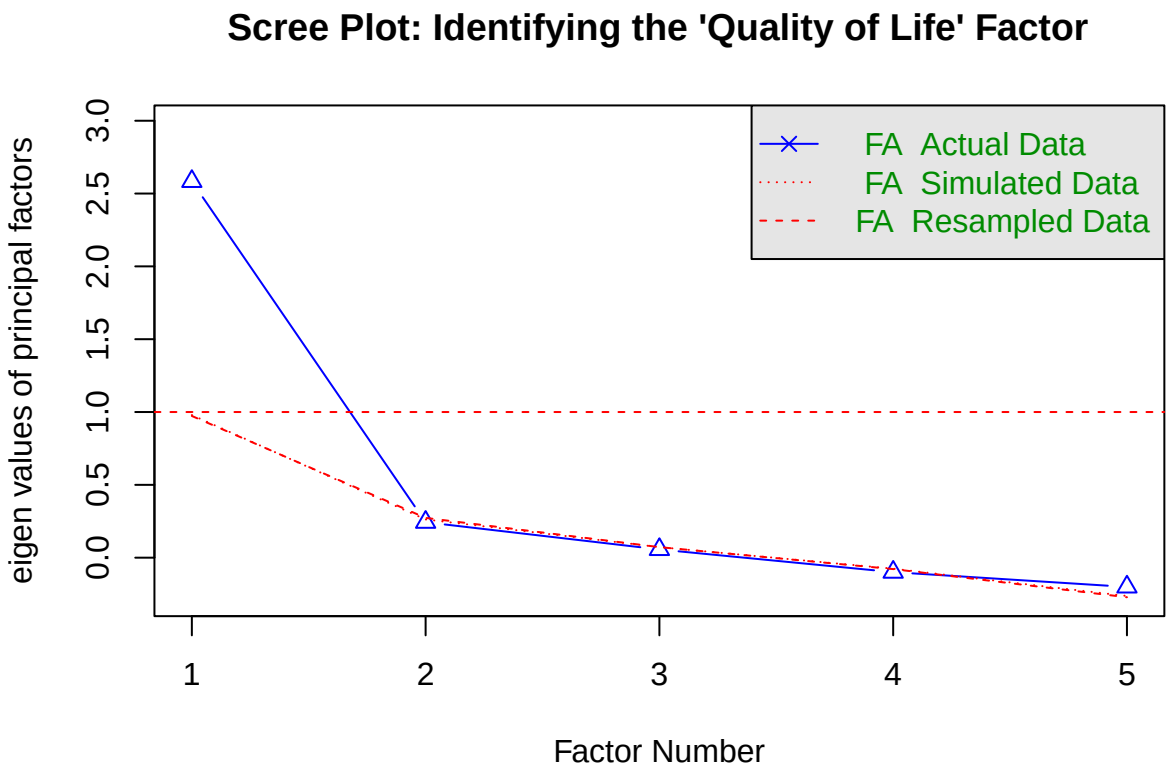

```r
# 2. Add a horizontal line at Eigenvalue = 1 (Kaiser Criterion)
abline(h = 1, col = "red", lty = 2)
```


Scree Plot: Identifying the 'Quality of Life' Factor
eigen values of principal factors
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1
2
3
4
5
Factor Number
FA Actual Data
FA Simulated Data
FA Resampled Data